

Regional medical resource synergistic security resilience assessment based on city network: A case study of YRD, PRD, and BTH

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ABSTRACT

Against the background of regional integration process and increasing urbanization level, the formation of highly integrated development regions like city clusters has greatly facilitated the medical resource synergy among cities, which significantly affects the regional medical resource security required for sustainable regional development and response to disaster attack. Improving regional resilience requires ensuring medical resources with a more rational spatial distribution and its synergistic development commensurate with economic and demographic development. Therefore, to further analysis medical resource security resilience of city cluster region, this study proposes a balanced approach including the perspectives of equilibrium of each city, the city network structure layout and recovery ability under disaster. With the case study of Yangtze River Delta (YRD), Pearl River Delta (PRD) and Beijing-Tianjin-Hebei (BTH), we comprehensively measure the regional medical resources security resilience from spatial distribution and temporal evolution perspectives. The results reveal that in spatial distribution perspective, the synergy intensity among cities in PRD is significantly highest, while in YRD is the most balanced. In addition, Beijing shows greatest medical impact in BTH. For temporal evolution perspectives, PRD and BTH show the tendency of medical resource synergy driven by core cities, while YRD create the multi-subscale synergies.

1. Introduction

Medical resource is an important basis for guaranteeing people's living standards and maintaining sustainable regional development. The ability to keep medical resource security is not only related to the development momentum of the region on a regular basis, but also an important barrier against external attacks, which has received widespread attention in recent years. Medical resource as a collective term for all types of factors of production occupied and consumed by regional governments in the provision of healthcare services (Xia et al., 2021), has a great guarantee for people's healthy life and life safety and is an important part of ensuring the normal and orderly operation of urban functions (Yu & Li, 2022). Nowadays, Increasingly frequent disaster and public health events under extreme climates are putting regional

development at higher risk (Monstadt & Schmidt, 2019). Medical resource security is also the basis to support the operations of the healthcare system and the delivery of emergency relief in these risks (Peng et al., 2022).

As the process of regional integration advances, regional inter-city links have become closer. The global world has gradually formed a number of regions with a high degree of integration of development, represented by metropolitan areas, city clusters, etc. (Kourtiti et al., 2020). These regions usually have one or more megacities as the core, and at least three or more large cities as the constituent units, relying on convenient infrastructure networks forming urban spatial clusters with compact spatial organization, close economic ties, and a high degree of integration (Fang, 2019; Fang & Yu, 2017).

Within these integrated city clusters area, the flow of medical

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resources is highly networked and coordinated among the cities (Kou & Zhao, 2024). The core cities have an overwhelming advantage over other cities in terms of policy, talent, economic augmentation and medical levels, which not only have to meet their own medical needs, but also those of their surrounding cities. According to People's Daily Health Client (<https://new.qq.com/rain/a/20230725A07LCK00>), in the construction of 125 national regional medical center projects, core cities such as Beijing and Shanghai have exported 50 and 20 projects respectively to the surrounding small and medium-sized cities, greatly promoting the regional medical balance. Meanwhile, the surrounding small cities also provide residents in core cities with more convenient daily medical services, effectively alleviating the medical congestion problems. According to Hebei Economic News (<http://www.hbjjrb.com/system/2024/03/01/101284584.shtml>), 14 counties (cities and districts) surrounding Beijing, also offers a wide range of basic medical services for residents of Beijing suburbs. As of 2023, 496,000 general outpatient visits were made by Beijingers to the 14 counties (districts) surrounding Beijing in Hebei, representing a year-on-year increase of 368.65 %, which also show the contribution of small cities in the region to the security of medical resources in the core cities. This regional inter-city synergy of medical resource relying on the city network has had a significant impact on the resilience of medical resource security. When faced with the risk of unexpected events such as disasters or public health events, regional synergies and mutual assistance of emergency medical resources based on the city networks are more evident (Peng et al., 2022). For example, when Wuhan was infected by Covid-19 in 2020, its own medical resources were in a state of near collapse. The entire cities in Middle Reach of Yangtze River City Clusters coordinated the deployment of medical resources to support Wuhan, which ultimately minimized the losses (Shan et al., 2023). It is necessary to revisit the issue of the regional medical resource security resilience from the perspective of urban synergy network in city clusters.

In addition, this cross-city synergy of medical activities will help to solve the unevenly distribution and inadequate supply of medical resources (Wang & Nie, 2024). For example, in 2007, only 750,000 Americans traveled abroad for medical care (Gan & Frederick, 2011), and this number increased to 1.4 million in 2017 (Dalen & Alpert, 2019), which means more people get health care and shows that the medical synergy has to some extent alleviated the problem of uneven distribution and inadequate supply. However, within different degree of regional integration and different development patterns, each city cluster shows difference in medical resource synergy security resilience (Wang & Nie, 2024). The quantitative evaluation and comparison of regional medical resource synergy security resilience among city clusters is of great practical significance in supporting sustainable development and guiding regional government administrators to allocate medical resources in a more rational manner.

Therefore, based on the regional medical resource synergy mechanism under the city network, this study quantitatively evaluates the medical resilience of three major city clusters of China in terms of spatial distribution and temporal evolution. We selected Yangtze River Delta (YRD), Pearl River Delta (PRD) and Beijing-Tianjin-Hebei (BTH) as case studies, and compared their resilience of medical resource synergistic security. The results reveal the differences of medical resources security resilience among the three city clusters, as well as the important city nodes and the weaker city nodes within each city cluster. This provides a theoretical basis for further improving the medical construction of the city clusters and the sustainable development of the region.

2. Literature review

2.1. Medical resource security

The scientific allocation and stability security of medical resources is related to whether residents can enjoy the various functions and services of urban medical resources conveniently and equally, which is an

important foundation for realizing the harmony and unity of the three pillars of economic development, social equity, and environmental friendliness (Ethel et al., 2017; Huang et al., 2021). In the past, the evaluation methods of medical resource security were mostly based on traditional databases to establish information systems and quantitatively evaluate medical resource allocation capabilities by using indicators such as the balance of medical resource supply and demand and the professional skills of medical staff (Belciug et al., 2020; Gong & Tang, 2020). Based on the collection of data, researchers constructed the indicator system to measure different region medical resources security capacity for regional improvement in medical industry. For example, Haakenstad et al. (Haakenstad et al., 2022) used comparable and standardized data sources to estimate human resources for health densities globally, quantifying human health resource scores and comparing the difference for 194 countries and territories. Ferrinho (Ferrinho et al., 2011) et al. described the way the human resources for health establishment is distributed in the different provinces of Zambia based on data on people working in health industry, with a view to assess the dimension of shortages and of imbalances in the distribution of medical resources by province and by level of care.

With the integration of GIS and medical resource security, researchers have further focused on the important impact of the spatial layout of medical resources on resource security. Spatial accessibility of medical resources has become an important dimension in assessing the ability of medical resources security (Zhao et al., 2021; Zhu et al., 2019). Yu (Yu, 2022) based on GIS spatial analysis methods, used the average nearest neighbor, kernel density estimation method, standard deviation ellipse method, and accessibility measure to analyze the spatial distribution of medical resources and its accessibility. In this type of spatial study of medical resource security, some researchers have found that the development status of the region and the accessibility of transport also significantly affects the security status of medical resources (Guo & Liang, 2022). Liu et al. (Liu, Gu, & Shi, 2022) constructed multiple modes of transportation (pedestrian systems, bus lines, rail lines) to evaluated the spatial accessibility and regional medical resources security from the perspective of service area ratio and service population ratio in the main urban area of Chongqing.

However, most of these studies have only considered the medical resource security capacity of individual regions or cities, neglecting the impact of the synergy and mutual assistance between cities under the discovery of regional integration. With the accelerated development of regional urban integration, the medical resources of a certain city will not only serve the city itself, but also serve the neighbors, which will have a significant impact on the overall medical resources security capacity of the whole region. Rational distribution of resources within the region will significantly increase the resilience of the whole region. Therefore, it is necessary to further consider the capacity of regional medical resources security under the synergistic mutual assistance of cities from the perspective of the region as a whole.

2.2. Region city synergy networks

Regions are never isolated; instead, they can affect each other (Overman et al., 2010). The rapid development of modern transport system and information industry makes the connection between the cities in the region closer, with the flow of people, information, and resources gradually building up a networked dynamic space between the cities (Wei & Xiu, 2020).

As shown in Table 1, in recent years, studies from spatial network perspective on regional inter-city synergistic relationships have widely appeared in various fields such as economy, ecology, infrastructure and social (Liu, Li, et al., 2022; Wang et al., 2023; Xie et al., 2020; Ye & Qian, 2021). Researchers consider cities as nodes and synergistic relationships between different cities as connecting edges, constructing different types of region city synergies network with entity or abstract data streams. Shi et al. (Shi et al., 2022) used a modified gravity model to construct an

Table. 1

Framework for resilience assessment of regional city networks.

Aspects	Study area of network	Reference	Network construction	
			Physical	Abstract
Economic	City relation network in Harbin-Dalian urban belt	(Xie et al., 2020)		✓
	Urban economic linkage network in Guanzhong Plain City Cluster	(Ye & Qian, 2021)		✓
	Supply chain network in Taiand	(Haraguchi & Lall, 2015)	✓	
Ecology	Chinese provinces city network	(Wang et al., 2023)		✓
	Global CO2 transfer network	(Zhang et al., 2017)	✓	
	Ecological Function Network in Taiwan Island	(Chang et al., 2007)		✓
Infrastructure	Global urban-industrial network	(Shughrue & Seto, 2018)		✓
	Infrastructure network of Brazilian city of Salvador's	(Silva et al., 2022)		✓
	Electricity transmission network in Netherlands	(Bollinger & Dijkema, 2016)	✓	
Social	Bonding and bridging networks on the Bangladeshi coast	(Islam & Walkerden, 2014)		✓
	Ecosystem network of the city of Sarno, Italy	(Bozza et al., 2017)		✓

economic synergy network, big data on user behavior to construct a social synergy network, and actual traffic data to construct an engineering synergy network, in order to further delineate the innovation synergistic relationships within the cities in Guangdong-Hong Kong-Macao greater bay area according to the above network model.

These studies can provide a methodological underpinning for the study of synergistic relationships of healthcare resources in regional dependent city networks. However, most of the above studies about the construction of regional city synergistic network model only focus on the single performance of the city and lack the characterization of its multifaceted complexity. The medical resources synergistic relationships among cities are constrained by the resources' spatial distribution, the accessibility of urban transport, and the level of economic and social development of cities. Its network is a more complex system. Most researchers are only aware of the existence of the complex relationship of regional medical resource synergy (Du et al., 2022), and there is still a gap in the modelling of its network. Therefore, it is necessary to further improve the complex modelling of regional medical resource synergy networks.

2.3. Measure regional resilience from network perspective

Whether the medical resources can meet the needs of the region under synergistic relationships and whether they can cope with the blows of sudden disasters determines the resilience of the region. Holling (Holling, 1973) first applied resilience theory to the field of systems ecology, and then the concept of resilience was gradually extended to regional systems (Macaskill & Guthrie, 2014). The regional resilience definition emphasizes the ability of a region to anticipate, prepare for, respond to and recover from a disturbance (Bristow, 2010).

The study of measuring regional network resilience started a little earlier, which have mostly measured regional network resilience from three perspectives. Several studies have focused on the node characteristics in the measurement of regional network resilience. Some network nodes play critical roles in connectivity and resource control

and are called influential nodes (Yang & Xiao, 2021). Identifying and improving these nodes is strategically important to facilitate network performance and enhance targeted preventive measures to reduce resilience loss from external shocks. Degree centrality, betweenness centrality, closeness centrality and k-shell are frequent measures (Dumedah & Garsonu, 2021; Ma et al., 2016), which can describe the node's own influence and its location influence for surrounding neighboring nodes. While for the whole region, the network structural is also not to be ignored. This is because the structure affects not only the efficiency and quality of factor flows but also how well the region responds to threats. The metrics such as network density, degree distribution, and degree correlation (Shen et al., 2021; Jiang & Ma, 2021), which are powerful tools for static evaluation and functional evaluation of network structures. Meanwhile, with risk strikes becoming more frequent, it is also meaningful to explore the regional resilience under dynamic strike perspectives (Xiong et al., 2022). This involves assessing the regional network resilience to node removal or connection disassociation through techniques such as disruption simulation (Kou & Zhao, 2024; Boschma, 2015).

However, most of the existing studies only have measured regional resilience in only one perspective. The regional medical resources security resilience should not only satisfy the optimum of normalized medical supply, but also meet the rapid recovery after disaster strikes. Its resilience capacity is constrained by the influence of the core cities in the region, the overall regional resource layout, and the recovery from unexpected strikes. Thus, measuring its resilience needs to incorporate multiple perspectives into the network analysis to enhance the comprehensiveness and refinement of the study. Furthermore, as regional resilience is presented in the context of regional spatiality and faces more complicated circumstances than the resilience of a single area, its development process is dynamic and evolving. The interpretation of the results of the resilience network analysis has significant implications for the planning direction.

2.4. Research gap

To sum up, medical resource security has great importance for sustainable regional development. With the urban integration development, the medical resource synergistic relationship of inter-city has a significant impact on regional resilience. However, the quantitative measurement on the medical resource synergy is still in a blank stage. The method of city network modelling can effectively describe the synergistic strength of inter-city, while existing studies only focus on the single performance of the city and lack the characterization of its multifaceted complexity. In addition, further resilience assessment studies based on networks lack measures that cover multiple aspects such as node characteristics, network structure, and dynamic strikes.

This study enriches the content of regional medical resource security resilience research in a network perspective, and the results can provide theoretical support for optimizing regional medical planning. The contributions of this study are as follows:

First, at the theoretical level, this study explains the medical resource synergistic mechanism between regional cities under the development of city networks.

Second, this study introduces a new methodology of constructing a regional synergistic network of city interactions from a relational perspective, which bridges the gap in quantitative research on cross-city medical resource security.

Third, this study evaluates regional medical resource security resilience from multiple perspectives of network analysis methods, enriching the methodological indicators of regional resilience study.

The rest of the article is organized as follows: Section 3 presents the data and method. Section 4 gives the results of the analysis of region medical resource security resilience of YRD, PRD, BTH. Section 5 reveals the implications of this study in guiding the sustainable development of medical security in the three major city clusters. Finally, in Section 6,

main conclusions, the contributions and shortcomings of this study are given.

3. Data and method

3.1. Study area and data resources

Nowadays, China is at a new stage of transformation and development of new urbanization and has entered a new era of leading the development of global city clusters in the 21st century. The experience and model of China's city cluster researches and construction are being emulated and learnt from by the global city cluster construction (Fang et al., 2018).

China's 13th Five-Year Plan identifies 19 key city clusters for development and planning, where YRD, PRD and BTH are the three with the most dynamic economies, the highest degree of openness, the strongest innovation capacity, and the largest number of inward absorbing populations (Zhang et al., 2021). In this study, we selected the above three city clusters as the study area to explore region medical resource security resilience and their development trend.

As shown in Fig. 1, YRD, PRD and BTH are all located in China's eastern coastal regions, gathering about one-third population and contributing nearly one-half GDP. The large scale and population expansion of these city clusters have also brought about certain problems of medical resources security during the development. Therefore, it is of great practical significance to analysis the medical resources security resilience for these three city clusters.

According to the existing literature shown in Table.2, the security of medical resources in a region usually involves both the supply and demand (Song et al., 2024). The supply of medical resources includes both material resources, such as hospital beds, and human resources, such as licensed (assistant) doctors. While the Demand for medical resource is usually proportional to the size of the population in the region. The larger the population, the greater the demand for medical resources. Meanwhile, the supply of medical resources to meet the demand is dependent on a number of factors such as easy access to transport, a well-developed economy, and population mobility.

This study obtains the corresponding medical level, population, GDP and other data of each region through China Statistical Yearbook,

Table. 2

Indicators of medical resource synergistic security in city clusters.

Dimensions	Attribute indicators	Reference
Supply for medical resources	Number of Beds of Hospitals Number of Licensed (Assistant) Doctors	(Zhang et al., 2016; Zheng et al., 2015)
Demand for medical resources	Numbers of Populations	(Dinh et al., 2022; Nádas et al., 2015)
Supply-demand linkage between cities	GDP Numbers of Populations Access Distance	(Wang et al., 2021)

Provincial Statistical Yearbook and China Urban Statistical Yearbook. In this study, multi-indicator panel data of years in 2010, 2015, 2020 are used as the basis for constructing the medical resources synergy networks in turn. Meanwhile, these period are selected for the following reasons. First, this is a crucial stage in China's new urbanization during the temporal evolution. As large numbers of people flock to city cluster regions, the problem of medical resources synergy and security has taken on a new dimension. Second, data for 2020 and beyond is unavailable for some indicators due to data availability.

3.2. Method of resilience analysis

As shown in Fig. 2, this section constructs an analytical framework for regional medical resource security resilience in city clusters. The first step is to calculate medical resource supply and demand in each city based on the improved location quotient model, and to establish the medical resource synergistic relationship among each city through the gravitational model, finally constructing the regional medical resource synergy network. Then, the second step is to carry out network analysis for the three dimensions of regional medical resource security resilience, including importance of city nodes in medical resource synergy, resilience of city network structural and resilience of recovery under attacks. Finally, according to making comparisons of regional medical resource security resilience among each city cluster in perspective of spatial distribution and temporal evolution, the current developing state and future improvement trends of each city cluster region will be revealed.

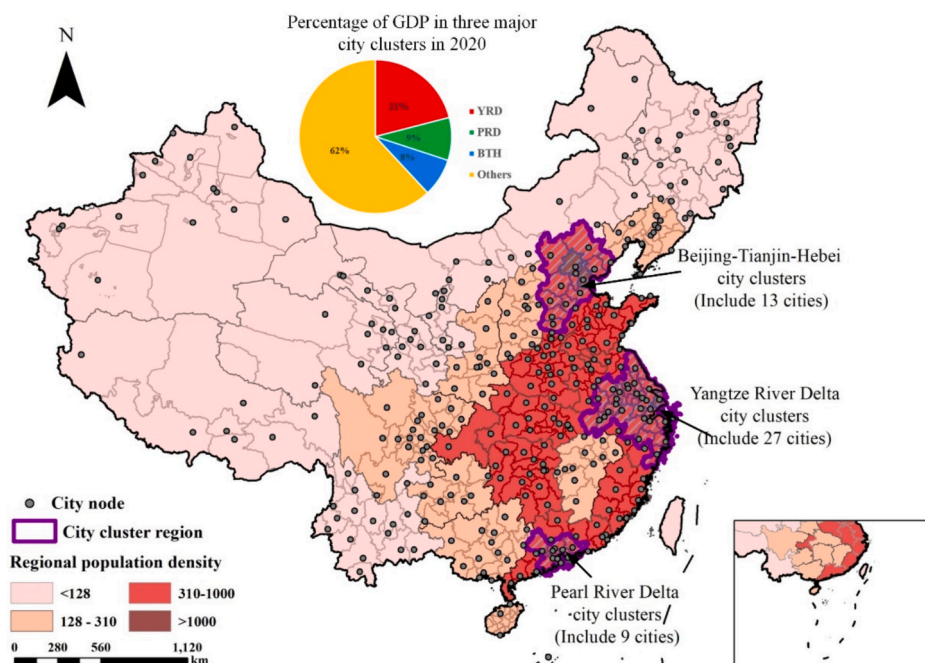


Fig. 1. Status of development of the three major city clusters.

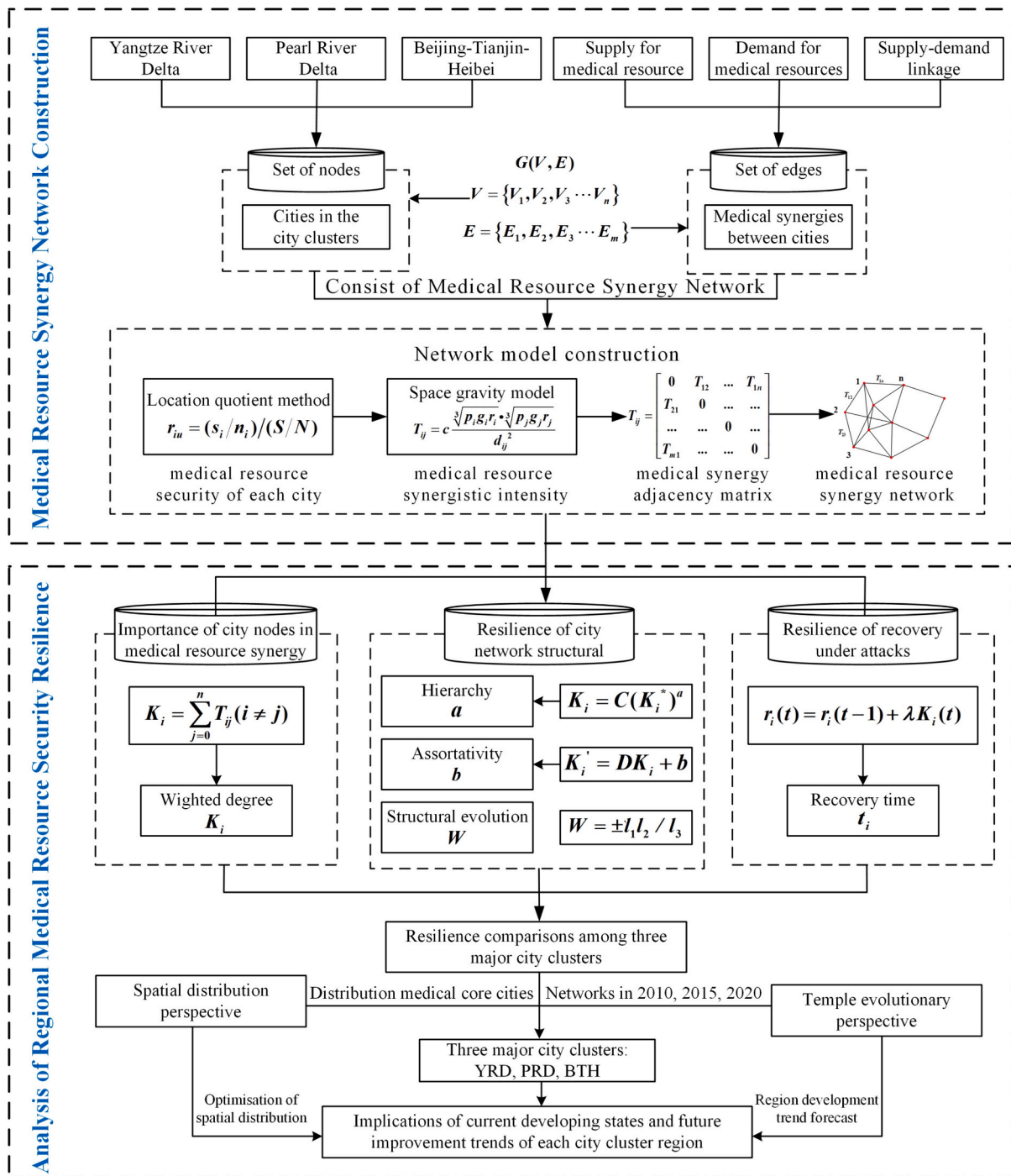


Fig. 2. Framework for analysis of regional medical resource security resilience.

The parameter definitions used in the analytical framework for regional medical resource security resilience are as shown in Table.3.

3.2.1. Medical resource synergy network construction

Medical resource synergy network $G(V, E)$ are formed when various medical resources represented by external medical security resilience functions circulate in the region network. The cities within the city cluster are selected as the set of network nodes $V = \{v_1, v_2, v_3, \dots, v_n\}$, and the synergistic relationship of medical resources between the cities is selected as the set of edges of the network $E = \{e_1, e_2, e_3, \dots, e_n\}$.

The construction of the regional medical resources synergy security network is a mechanism of connection based on the security of medical

supply and demand of each city in the region, based on the strength of the synergistic relationship between the cities. Based on their own supply and demand of medical resources, each city is spatially linked with other cities in the city cluster region, so as to meet medical resource synergy security.

> Supply and demand security of each own city

The location quotient method, also known as the rate of specialization, is a method to calculate the superiority and inferiority of an industrial sector in a region (Fracasso & Vittucci, 2018). Based on the traditional location quotient method, this study proposes an improved

Table. 3

Parameter definition used in the resilience assessment framework.

Parameter	Definition
$G(V, E)$	Medical resource synergy network
$V = \{v_1, v_2, v_3, \dots, v_n\}$	The set of network nodes
$E = \{e_1, e_2, e_3, \dots, e_n\}$	The set of edges of the network
r_i	The regional resilience quotient of medical supply and demand
r'_i	The regional resilience quotient of medical supply and demand after attack
s_i, S	The medical resource supply of city i (of the whole country)
n_i, N	The medical resource demand of city i (of the whole country)
$s_i(1), S(1)$	The medical supply of the number of hospital beds
$s_i(2), S(2)$	The medical supply of the number of hospital beds
p_i, P	Population of city i (of the whole country)
g_i	GDP of city i
T_{ij}	The synergistic intensity T_{ij} between city i and city j
d_{ij}	The transport distance between city i and city j
K_i	Weighted degree of city node i
K_i^*	The rank of city node i in the degree distribution
K_h	Weighted degree of city node h which is neighboring node of node i
K'_i	The average weighted degree of neighboring nodes of node i
a	Hierarchy character of network structural
b	Assortativity character of network structural
W	Resilience evolution character of network structure
t_i	Recovery time of city i after disaster attack

method for calculating the location quotient of medical resources security resilience of each city within a city cluster. By dividing the medical supply and demand of the cities themselves with the whole country, the aims are to eliminate the factor of regional scale differences and reflect more accurately the levels of medical supply and demand resilience in each city under a unified standard with the following Eq. 1.

$$r_i = (s_i/n_i)/(S/N) \quad (i = 1, 2, \dots, n, u = 1, 2, \dots, m) \quad (1)$$

s_i and n_i on the numerator respectively represent the medical resource supply and demand in the region for city i itself. Accordingly, S and N in the denominator represent the medical resource supply and demand in the whole country. By comparing the supply and demand conditions, the regional resilience quotient r_i of the medical resource security level of city i is then determined.

As shown in Eqs. 2 & 3, the medical resource supply in this study consists of both the number of hospital beds ($s_i(1), S(1)$) and the number of professional doctors ($s_i(2), S(2)$), while the medical resource demand is proportional to the size of the corresponding population (p_i, P). Where ρ, σ, k are the corresponding constants

$$s_i = \rho s_i(1) + \sigma s_i(2); S = \rho S(1) + \sigma S(2) \quad (2)$$

$$n_i = k p_i; N = k P \quad (3)$$

➤ Quantitatively calculate medical resource synergistic relationship

The quantitative calculation of the medical resource synergistic relationship between cities is expressed as the synergistic intensity T_{ij} . The spatial gravity model, which is based on the principle of spatial interaction, can represents the strength of medical resource synergy relationship between cities (Du et al., 2023; Wang et al., 2023). The calculation of synergistic intensity T_{ij} is shown in Eq. 4, which shows the medical resource synergistic strength between cities is positively related to the economy, medical supply and demand, population mobility, and negatively related to the spatial distance.

$$T_{ij} = c \frac{\sqrt[3]{p_i g_i r_i} \sqrt[3]{p_j g_j r_j}}{d_{ij}^2} \quad (4)$$

c is the gravity coefficient, usually taken as 1. p is the population size, g is the GDP of the city and d_{ij} is the transport distance between city i and city j .

The calculation results of T_{ij} leads to the construction of the medical resource security network adjacency matrix. Meanwhile, to facilitate the characterization of the network, binarization is applied in this study. The median is chosen as the threshold value, and when the synergistic intensity T_{ij} between the two provinces is greater than the median of the corresponding column elements, it is recorded as the edge exists, otherwise, it does not exist.

3.2.2. Analysis of regional medical resource security resilience

Analysis of medical resources synergy network is an important basis for measuring the medical security level of whole region and its ability to resist risk attacks. Based on the network analysis method, the regional resilience ability of medical resource security from multiple dimensions is explored.

(1) Importance of city nodes in medical resource synergy

City cluster region is composed of different levels of cities, and the synergistic effect of core cities and small cities on the regional medical resource security varies. Identifying cities with higher medical resource synergy importance will help to further improve the corresponding disaster defense and regional security.

According to the different resilience importance of each city node in synergistic linkages, we propose the concept of network weight degree based on traditional degree to reflect the importance of core city nodes in the network. Traditional degree quantifies the number of edges connected to a node, which mainly includes the number of nodes that lead to that node and induced by that node. While, for the medical resource synergy network, the synergistic intensity T_{ij} varies among cities, so we use the weight degree K_i to measure the importance of each city node in the medical resource synergy network.

$$K_i = \sum_{j=0}^{j=n} T_{ij} (i \neq j) \quad (5)$$

(2) Resilience of city network structural

Resilience analysis based on the city network structure can reflect the strength of synergistic relationship among cities. In this section, we further characterize the regional medical resource security resilience under the perspective of city network structure in terms of hierarchy, assortativity, and structure evolution.

➤ Hierarchy

Hierarchy represents the weighting ranking of nodes in a network. A more hierarchically organized city network means that high degree cities are more pronounced and conducive to efficiency and competitiveness, which prevents lock-out. But when high degree cities suffer from any huge shocks, the network's vulnerability increases. In contrast, a less hierarchically organized city network means that cities in the network are less sensitive to random node failure. Removing a certain node has little effect on the whole network (Ye & Qian, 2021). Scholars often use the value of the slope a of the degree distribution to express the hierarchical capacity of the network. The degree distribution usually uses the following Eq. 6 to quantify hierarchy.

$$K_i = C(K_i^*)^a \quad (6)$$

K_i^* is the rank of node i in the degree distribution. It is usually treated on a log-log scale as follows:

$$\ln(K_i) = \ln(C) + a \ln(K_i^*) \quad (7)$$

C is a constant; a is negative, indicating the slope of the degree distribution curve. The larger the value of the slope a , the higher the hierarchical nature of the network, the more prominent the core position of the high value nodes. In contrast, the smaller the a value, the more reasonable the degree of external connectivity of the network nodes, meaning that the network presents a flattened development.

➤ Assortativity

The property of assortativity describes the connection relationship between nodes in the network by calculating the degree correlation (Newman, 2002). The correlations between nodes in a network are either assortative or disassortative (Newman, 2003). The network is assortative if nodes with large (or small) degrees tend to connect nodes with large (or small) degrees; otherwise, the network is disassortative.

First, for each node i , calculate the average weighted degree of its neighboring nodes as shown in Eq. 8:

$$K'_i = \sum_{h \in V_i} K_h / K_i \quad (8)$$

V_i is the set of neighboring nodes and K_h is the degree of neighboring node h of node i , and for estimating the linear relationship between K_i and K'_i .

$$K'_i = DK_i + b \quad (9)$$

D is a constant. When $b > 0$, the network is assortative, which means that city nodes with higher weight values are able to maintain good interaction with nodes at the same level as well as communicate and co-operate with city nodes at different levels; when $b < 0$, the network is disassortative, which means that linkages between core and peripheral urban nodes are relatively weak and cross-regional exchanges and cooperation in factor flows are limited.

➤ Structure evolution

Based on the changes in the 2 metrics a and b of hierarchy and assortativity under time evolution, the level of enhancement (decline) of network structural resilience trend can be preliminarily quantified. Fig. 3 illustrates the comprehensive network structure resilience development assessment model proposed by Xie et al. (Xie et al., 2020). The

point $(\Delta|a_1|, \Delta b_1)$ reflects the evolution status of the hierarchy and assortativity of regional medical resource synergistic security network structure over time. If the point $(\Delta|a_1|, \Delta b_1)$ falls in the upper half of the straight line $\Delta b = \Delta|a|$, it indicates that the network structural resilience has been weakened during the evolution process. While it falls in the lower half of the straight line $\Delta b = \Delta|a|$, it indicates the enhanced trend of resilience. In addition, the distance l_1 between the point $(\Delta|a_1|, \Delta b_1)$ and the coordinate origin indicates the magnitude of the change of the structural resilience of the network, the larger l_1 is, the larger the magnitude of the structural resilience evolution is. The ratio of the perpendicular distances l_2 of the point $(\Delta|a_1|, \Delta b_1)$ from the straight line $\Delta b = \Delta|a|$ and the perpendicular distance l_3 from the straight line $\Delta b = -\Delta|a|$ reflects the quality of structural resilience evolution. A larger l_2/l_3 indicates a better quality of network structure resilience development or a more direct resilience decline. Accordingly, we constructed a network structure resilience synthesis— $W = \pm l_1 l_2/l_3$.

(3) Resilience of recovery under attacks.

The resilience of regional medical resource security is also reflected in its ability to recover from external attacks such as disasters and public health events. Based on the dynamic perspective, the study carried out the simulation of disruption and recovery after the city nodes suffered from risk attacks, transferring the city from static topology to destructive dynamic process to explore the regional medical resources security resilience.

When a city in the region is perturbed by external attacks, its own medical security will be significantly reduced. As shown in Eq. 10, and we define the city's resilience quotient after attacks as r'_i , and the impact coefficient of the risk attack is defined as η .

$$r_i(t=0) = r'_i = \eta r_i \quad (0 < \eta < 1) \quad (10)$$

At this time, under the resource deployment of medical resource synergy network, the stricken city's resilience quotient will gradually complete the recovery at time t_i as shown in Eqs. 11 & 12, Resulting in a resilience indicator t_i for the city in the medical resource synergy network.

$$r_i(t) = r_i(t-1) + \lambda K_i(t) = r_i(t-1) + \lambda \sum_{j=0}^{j=n} T_{ij}(t) \quad (11)$$

$$r_i(t_i) = r_i(t=0) \quad (12)$$

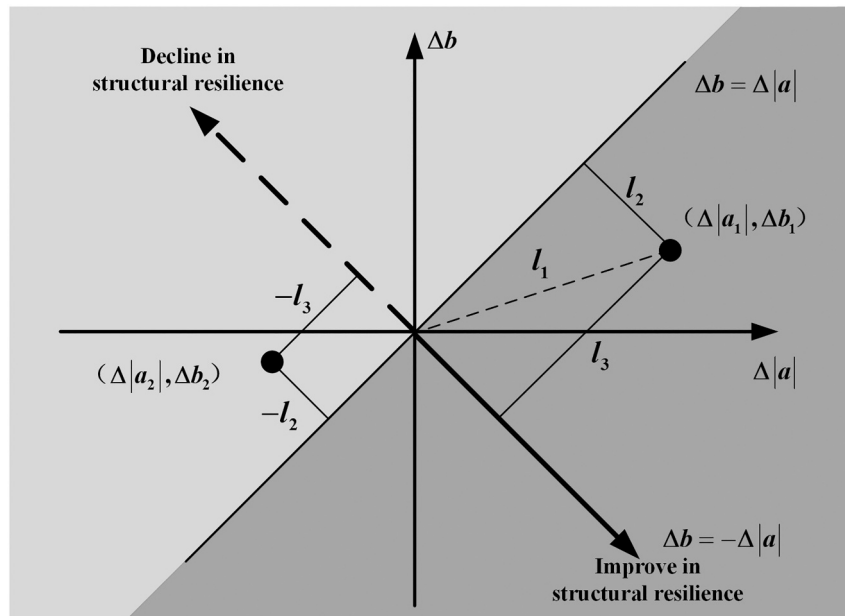


Fig. 3. Evolution of network structural resilience.

The smaller t_i means the stronger in the dimension of recovery resilience of the city i will be.

4. Results

4.1. Construction of medical resource synergy network in the three major city clusters

According to the calculation results of medical resource synergistic relationship, edges with a strength less than the median are excluded and classified into 5 levels. The visual representation of medical resource synergy networks using ArcGIS as shown in Fig. 4.

The overall spatial distribution within each city cluster illustrates that there are some differences in the medical resilience location quotient which represent the satisfaction of supply and demand in each city, but the impact on the whole regional resilience under medical resource synergy is not significant. While, the decay of geographical distance significantly affects the strength of medical resources synergistic security resilience. The majority of network edges are low synergistic intensity levels at Level I&II, while network edges with higher synergistic intensity levels at level IV&V are mostly concentrated between large cities in close spatial proximity such as Guangzhou and Foshan. The higher the transport accessibility to the other cities in the cluster, the more important resilience impact the corresponding city nodes are in the network.

In addition, in terms of temporal evolution, the medical resource synergistic intensity edge between cities in all three city clusters has increased over time. Compared with the 2010–2015 time period, the increase in the medical resource synergy intensity in the 2015–2020 time period is more obvious in all city clusters, indicating that medical synergy among city clusters develops rapidly and the interaction of medical resources is further strengthened in this period. The comparison among the three city clusters also shows that the medical resource synergy intensity of network edge in PRD is much higher than two other city clusters. Moreover, PRD and YRD both have a more pronounced increase than BTH over time.

Table 4 demonstrates the basic characteristics of regional medical resource synergistic network in the three city clusters. In PRD, the synergy intensity of medical resources among cities is relatively high, while the gap of the synergy intensity of medical resources among cities in the YRD and BTH is much smaller.

4.2. Analysis of medical resource security resilience in the three major city clusters

4.2.1. Importance of city nodes in medical resource synergy

According to the Eq. 4, all the city nodes' weighted degree are calculated to show the difference of resilience impact on the whole medical resource synergy. Fig. 5(a) & (b) demonstrates the weighted degree of each city nodes in the corresponding city cluster. Meanwhile,

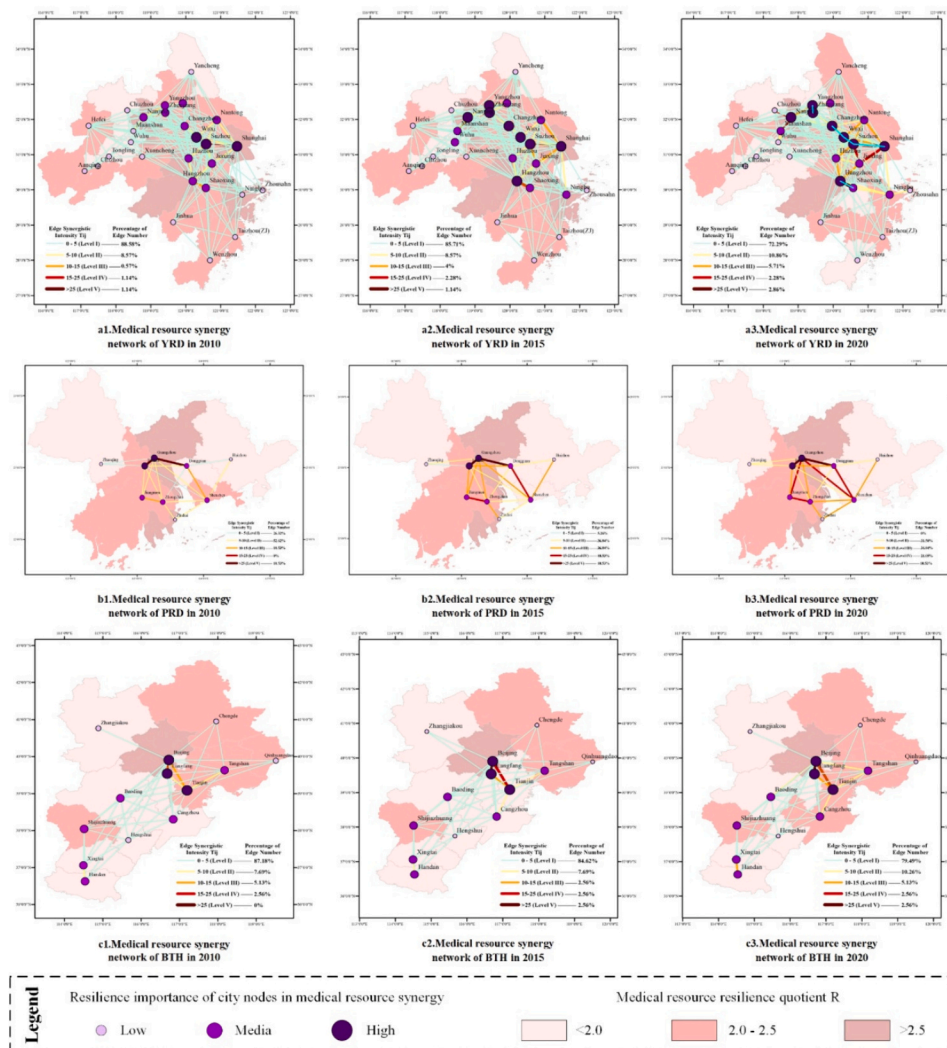


Fig. 4. Medical resource synergy network in the three major city clusters.

Table. 4
Network characteristics.

Region Network	YRD			PRD			BTH		
Number of nodes	27			9			14		
Number of edges	175			19			39		
Years	2010	2015	2020	2010	2015	2020	2010	2015	2020
Mean of T_{ij}	1.19	1.64	2.31	9.74	13.00	17.50	1.21	1.82	2.18
Median of T_{ij}	0.34	0.48	0.66	3.72	5.03	6.46	0.37	0.52	0.60
Max of T_{ij}	31.02	43.50	51.82	182.66	238.09	322.09	20.30	28.19	39.22
	(Yangzhou—Zhenjiang)			(Guangzhou—Foshan)			(Beijing—Langfang)		
Std. of each node's weighted degree	4.98	6.47	9.37	29.55	38.39	52.02	3.86	5.62	6.99

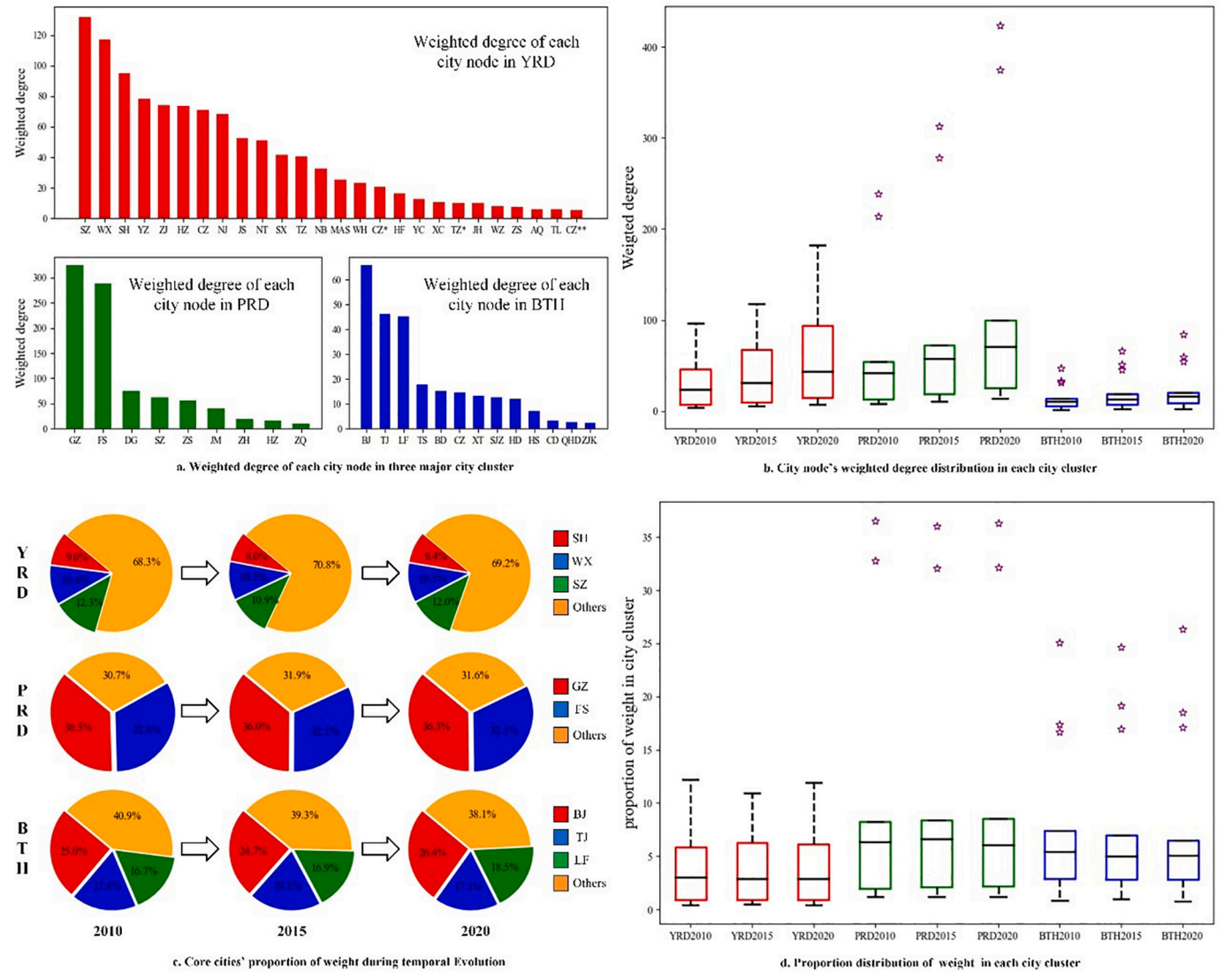


Fig. 5. Weighted degree of city nodes in each network.

Fig. 5(c) & (d) reveal the city's proportion of weighted degree after normalizing. In YRD, Shanghai, Wuxi, and Suzhou are the core cities with the highest weighed degrees in the medical resource synergy network, while their influence in the medical resource synergy is not prominent and there are relatively smaller gaps of weighted degree

among cities. However, in PRD, Guangzhou and Foshan are in the clear dominant position of medical resource synergy which have the greatest impact on resilience, and the sum of the weight share of the two cities exceeds 50 %. Similarly, in BTH, the intensity of medical resource synergy between Beijing, Tianjin, and Langfang is in the perspective of the

dominant position of these three cities, which has a tendency to be further strengthened with the development of time.

4.2.2. Resilience of city network structural

(1) Hierarchical

The degree distribution regression coefficient $|a|$ of the hierarchical index was calculated according to Eqs. 5–6. The results are shown in Fig. 6. The degree distribution regression coefficient $|a|$ of PRD city cluster network has the largest value, which indicates that its hierarchical nature is most obvious. The core cities within PRD such as Guangzhou and Foshan are prominent, but there may be a path dependence defect. The degree distribution regression coefficient $|a|$ value is the smallest in YRD city cluster network, indicating a relatively flat and stable network topology, with a relatively small gap in the degree of weight between cities, and a relatively homogeneous medical resources among cities without many powerful dominant nodes.

With the economic development and population surge under the time evolution, the degree distribution coefficient $|a|$ of each city cluster network all have changes. The hierarchical nature of the YRD city cluster network gradually decreases, and the synergy of medical resources among cities becomes more and more balanced, while the degree distribution regression coefficient $|a|$ of BTH city cluster network gradually becomes larger, showing a more obvious dependence on the core cities.

(2) Assortativity

According to Eqs. 7 & 8, the results of the network structure assortativity index b are shown in Fig. 7. The values of degree correlation b of the three city clusters are all <0 , which shows the heterogeneity of the network structure and the city with a larger weighted degree connected to the city with a smaller weighted degree. The results of the comparison of the three city clusters show that BTH has the strongest heterogeneity in the medical resource synergy network, with a more diversified form of network path connection. There is also a better connection between core and non-core cities, and Beijing and Tianjin have better radiation ability to the surrounding area. On the other hand, the heterogeneity of the network structure of the PRD city cluster is weaker, which is mainly manifested in the strong connection of healthcare resources between the core cities of Guangzhou, Foshan, and Shenzhen, and the weak correlation of nodes in cities with less resilience.

On the time evolution scale, the degree of heterogeneity of the regional medical resource synergy network structure of the three city clusters all have decreased, indicating that on the city cluster scale, the synergy of medical resources has developed towards mutual assistance among core cities, and the connectivity between core cities with high weighting has been further strengthened.

(3) Structural evolution

Fig. 8 demonstrates the time evolution of each medical resources synergy network structural resilience in the three city clusters over the two time intervals 2010–2015 and 2015–2020. Overall, all coordinate points in each time intervals all fall in the upper triangular part of the coordinate system, which mean a decreasing trend in network structural

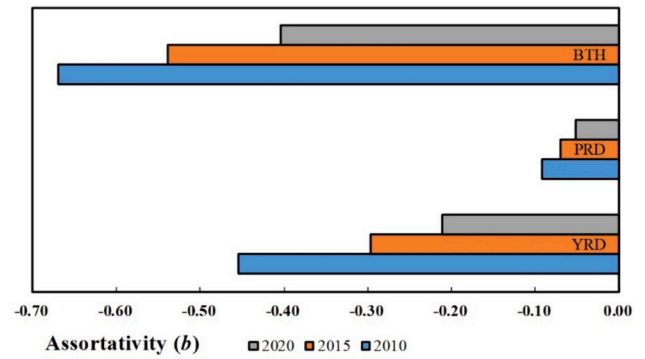


Fig. 7. Assortativity of different city cluster network resilience of each stages.

resilience. According to the calculation of network structure resilience evolution synthesis ($W = \pm l_1 l_2 / l_3$), YRD and PRD show the most significant decline in the 2010–2015 period. In addition, compared with the 2010–2015 period, the decline in network structural resilience of three city clusters in the 2015–2020 period all have slowed down, which reflects that the implementation of the corresponding measures has begun to effectively alleviate the pressure on medical resources security.

4.2.3. Resilience of recovery under attacks

This study simulated the recovery abilities of the medical resource synergistic security network constructed for the three city clusters of YRD, PRD, and BTH from the disruption. Assuming that each city in the cluster suffers an attack to disruption, the recovery process is simulated after the damage is achieved with the mutual support of the medical resource synergy network. During the experiment, we gave the following hypothesis descriptions for the disaster attack scenario:

- There is a single attack event—we assume that there is only a single impact on the network at the initial moment, and no other types of attacks are introduced in the process of urban medical resources security.
- The impacts of each city disruptions are all fixed—we assume that when a city within a regional city cluster is under the state of disruption after the attack, the destruction factor η is 50 %, which means that the city's medical resource security level would drop to 50 % of its initial capacity.
- All cities will reach full recovery from disruption—Within a certain period after the attack, all cities will eventually return to the normal state relying on the medical resource synergy network of co-giving. Recovery time t_i varies across cities, reflecting variability in the regional medical resources security resilience.

Fig. 9 (a), (b), and (c) show in turn the process of disruption to recovery for each city under the operation of 2020 medical resources synergy network in YRD, PRD, and BTH regions. The results show that

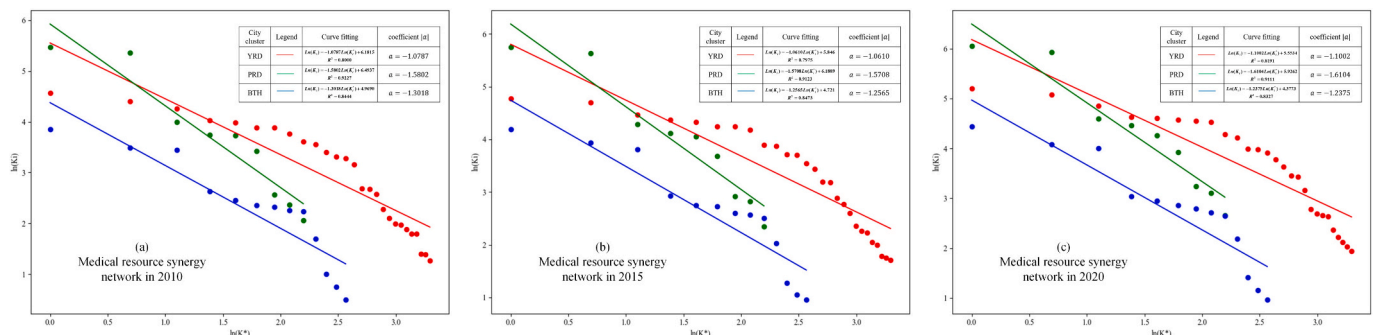


Fig. 6. Degree distribution of different city cluster network resilience of each stages.

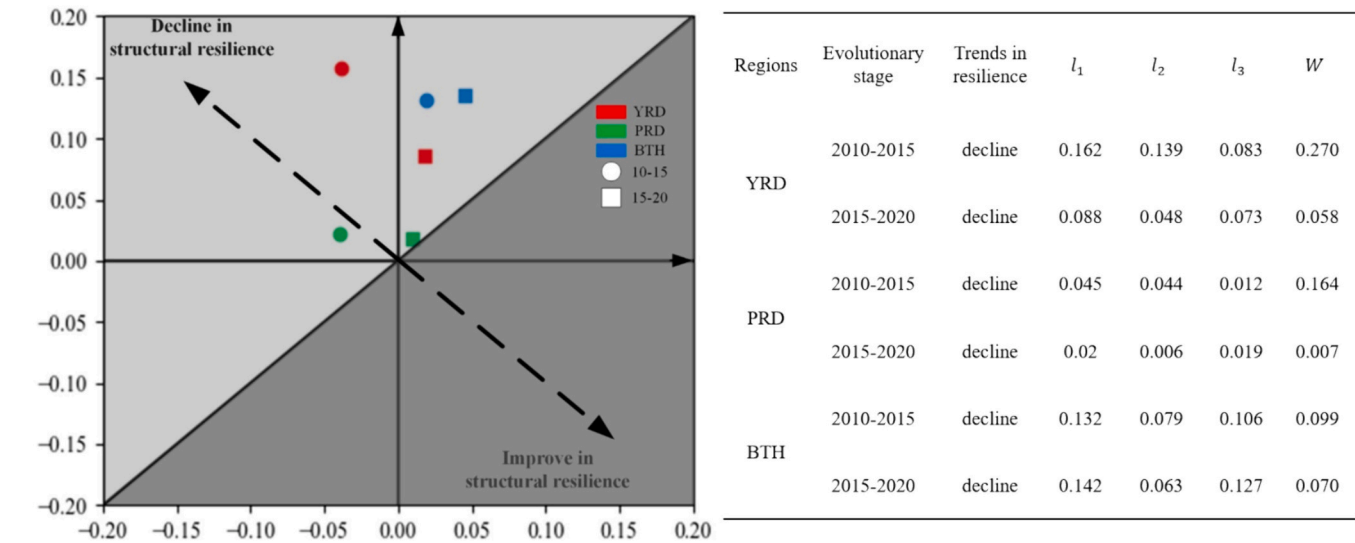


Fig. 8. Analysis of the time evolution of the structural resilience of city clusters.

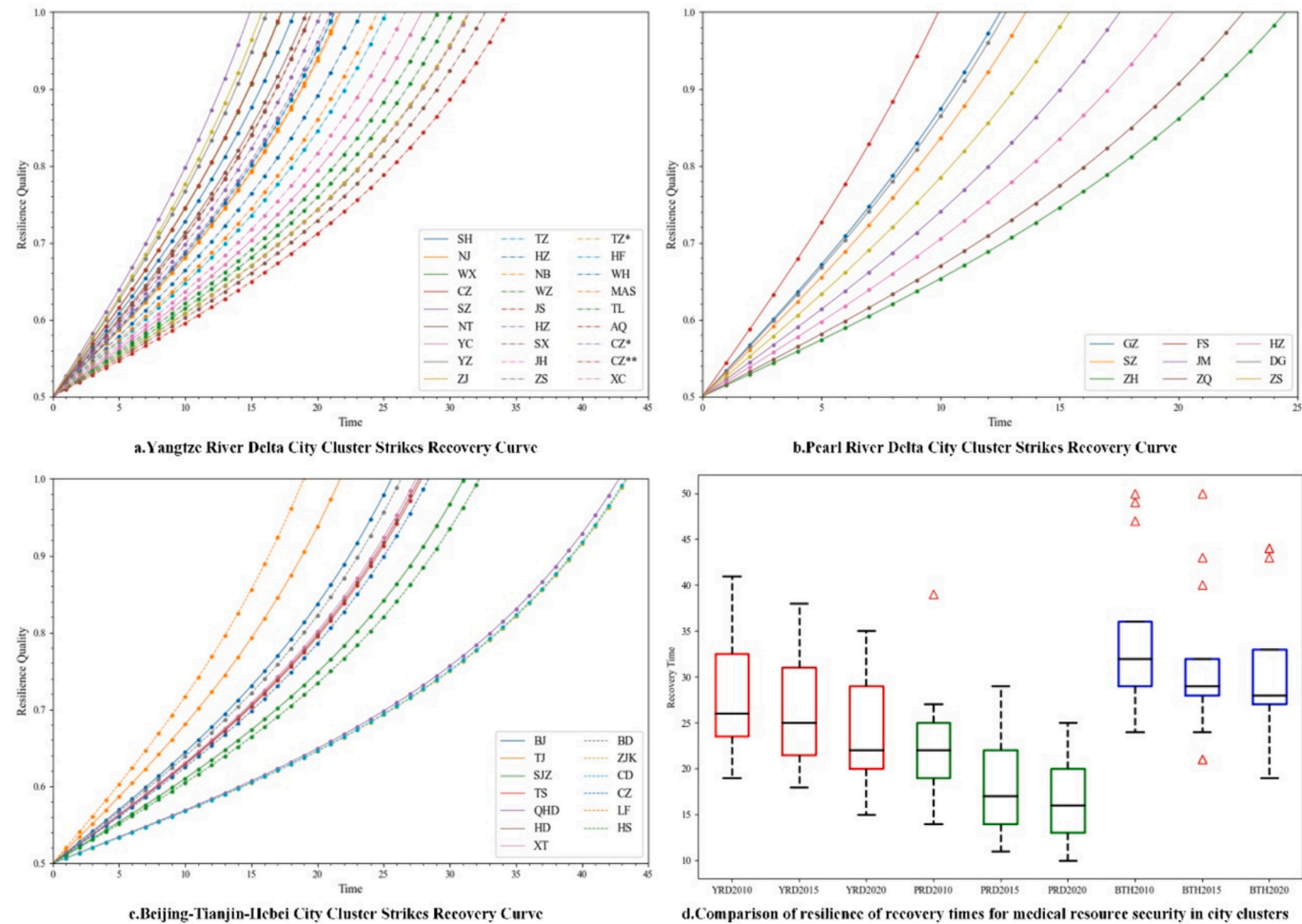


Fig. 9. Resilience recovery of medical resource security in three major city clusters attacked by disasters.

Suzhou, Foshan and Langfang are the cities with best recovery ability in each cluster, while Anqing, Zhuhai and Chengde have relatively weak recovery ability.

Fig. 9(d) comparatively demonstrates the recovery time of each city in the three city clusters after the attack relying on the regional medical

resources synergy network. Cities in PRD are relatively more capable of recovering from external shocks, while cities in BTH is relatively less capable of recovering from external shocks, and Zhangjiakou, Chengde, and Qinhuangdao consistently have abnormally long recovery times. In addition, in the comparative analyses of 2010, 2015 and 2020, the

recovery time of each city in all city clusters decreases, demonstrating the improvement of recovery ability in resilience. The decreasing slopes of the average recovery time are greater in YRD and PRD, demonstrating a more significant increase in the recovery capacity in the resilience of the two regions.

4.3. Comparison of resilience among each city cluster

Spatial distribution perspective and temporal evolution perspective have always been the focus of attention in the region studies (Ju, 2023; Zhou et al., 2021). This section makes comparisons of regional resilience among city cluster regions from the perspective of spatial distribution and temporal evolution,

4.3.1. Spatial distribution perspective

The analysis results of regional medical resource security resilience among the three major city clusters show that the synergy intensity among cities within PRD is at a high level, significantly higher than that of YRD and BTH. While, the variability of the medical resources synergy among individual cities within PRD is relatively large. Specifically, Guangzhou and Foshan have the most prominent synergy intensity, which is highly consistent with the integration development of the two cities as revealed by relevant studies in recent years (Wu et al., 2020). In contrast, cities such as Zhaoqing, which is at the edge of PRD, show a significant lag in medical resource synergy with other cities. It is not difficult to see that the overall medical emergency development in the PRD shows a difference in strength with Guangzhou as the core and gradually decreasing outwards.

In addition, the overall medical resource synergy intensity between cities within BTH and YRD is close in intensity, while there are still two different development models that are lacking. BTH shows a more obvious bifurcation, with Beijing, Tianjin and Langfang as the core cities with relatively high values of medical resource synergistic security, while the other cities have relatively low values. On the other hand, YRD shows more of a multiple core sub-network group with Shanghai-Suzhou-Wuxi, Nanjing-Yangzhou-Zhenjiang, and Hangzhou-Shaoxing-Jiaxing. The developmental differences in medical resources synergistic security network between the two city clusters may be due to the influence of national strategies as well as synergies in economic development. BTH relies mainly on the dual core drive of Beijing and Tianjin to drive the development of the surrounding cities. The GDP of Beijing and Tianjin is also significantly higher than that of other cities in the cluster. Langfang, as a bridge between Beijing and Tianjin, receives a large amount of medical radiation from the two places, and its ability to keep medical resources security is also at a high level. However, YRD Internally includes several developed metropolitan areas, such as Shanghai, Nanjing and Hangzhou, and their medical resources are also distributed in multiple core sub-networks.

4.3.2. Temporal evolutionary perspective

During the evolution state of 2010, 2015 and 2020, the medical resource security resilience of the three major city clusters shows different development trends. The development status of the weight share of core medical cities shows differences. As for PRD and BTH, the influence of regional medical synergy of the core city is gradually expanding, while YRD's medical synergy of the weight share of the core city of the fluctuation of the smaller. This indicates that the development of medical standards in the PRD and BTH is moving towards centrality, while the development of the YRD region is more balanced.

Comparing the changes in the network structure of the three city clusters over time evolution, the trend of declining structural resilience development is shown in all of them. And relative to 2010–2015, the declining trend of network structural resilience slows down during 2015–2020. As shown in Table 5, this declining trend of network structural resilience about medical resource security is correlated with the population growth trend in city cluster. The population of the three

city clusters all show strong growth, which greatly exacerbates the regional pressure on the regional medical resource synergy security. With the development of urbanization to a certain extent, the population growth rate has gradually slowed down, and the pressure of regional medical resource synergy security has been reduced to a certain extent.

Furthermore, the three major city clusters show a more positive trend in resilience in the face of disaster attacks. Fig. 9(d) shows that the resilience of each city in terms of medical resource synergistic security resilience is gradually improving as time evolves. With the increasing frequency of sudden-onset disasters and public health events in the face of global climate change, such an increase in medical resource synergistic security resilience will be an important positive signal.

5. Implications

From the above discussion, we summarize several important implications about regional medical resource in these city clusters, which are of significant guidance for maintaining sustainable development.

1. The regional medical resource synergy strength between in-cities in PRD is the best, while in YRD is the most balanced.

The comparison results of medical resource security resilience among the three major city clusters show that PRD has the best resilience with the highest degree of synergy and the best recovery ability. This may be related to the close integration between cities and the relatively close travelling distance. According to data from AMAP, the average travelling distance between cities in PRD is 89.78 km, much smaller than the 241.57 km in YRD and the 259.74 km in BTH. A more convenient transport will further strengthen the connection between cities within the city cluster region and facilitates medical resource synergy security. While, the difference of medical resource security resilience within YRD is relatively small, and its development also presents a more balanced development model. As of 2024, the YRD includes four national level metropolitan areas: Shanghai, Nanjing, Hangzhou, and Hefei. Almost all cities are located within the corresponding metropolitan areas, which promotes their balanced development. This balanced model of medical resource synergy security will be more conducive to the YRD region's ability to cope with unexpected risks, as compared to BTH and PRD, where the impact on the entire region in the event of damage to one of core cities is likely to be relatively small.

2. Beijing has an absolute influence in the medical resource synergy security of BTH among three city clusters.

From the medical resource synergy network in the three city clusters (as shown in Fig. 4), we find that Beijing is the only city in three clusters that has a contiguous relationship with all the cities. Meanwhile, in the results of measuring the resilience of each city in the BTH city cluster, weighted degree of Beijing is twice as important as that of the second-place Tianjin, reflecting the absolute importance of Beijing for medical resource synergy in BTH. While for YRD and PRD, the influence of the core cities is not as significant as Beijing in BTH. This is closely related to Beijing's unique political influence. As the political center of China, Beijing owns the highest number of tertiary hospitals in the country has received the most policy preferences. Beijing's influence in the medical sector is crucial. It is responding to, strengthening the policy orientation will help to enhance the ability of core cities to medical resource synergy

Table. 5
Comparisons of Network structural evolution trend and population growth trend.

City cluster region	Evolutionary stage	Network structure evolution trend W	Population growth trend
YRD	2010–2015	−0.2697	0.2214
	2015–2020	−0.0577	0.1451
PRD	2010–2015	−0.1640	0.3090
	2015–2020	−0.0065	0.3888
BTH	2010–2015	−0.0985	0.1587
	2015–2020	−0.0704	0.0571

security and increase their regional influence in the city clusters. However, once Beijing is hit by the risk of medical shortages, the whole BTH region will be affected by a huge ripple effect. This further illustrates the necessity of the policy of Beijing's Non-capital Function Redistribution (Wang & Yang, 2020).

3. PRD and BTH are further strengthening medical resource synergistic intensity between core cities, while YRD is trying to create multi-subscale medical synergies.

City clusters with different characteristics have realised different development paths to improve the regional medical resource resilience. In BTH and PRD, the core leading cities within the city cluster have better development prospects and higher leading weights, so it would be more valuable to further increase the intensity of medical resource synergy between the core cities to lead the development of the entire city cluster. While, for regionals such as YRD, which has a large number of cities and a larger land area, advocating the establishment of multi-subscale medical resources synergies and the establishment of multiple medical resource security cores is of greater significance for the regional medical resource security resilience.

4. The increase in the population of city clusters has intensified the pressure on the medical resource security, while the decline in the growth rate is gradually slowed down the pressure growth rate.

As shown in Table 5, the decline in the medical resource resilience coverage in each city cluster is correlated with the population growth trends over the time evolution. The rapid population growth in the three regions during the 2010–2015 led to a significant decline in network resilience. And with the gradual slowdown in population growth during the 2015–2020, the pressure on the region's medical resource security has been reduced to some extent. It can be expected that regions of city clusters with population growth in the future will further expand their demand for medical resources, exacerbating the pressure on medical resource security resilience. While, with the emergence of the trend of reverse urbanization. The core cities in the city cluster will experience population spillover, and the medical resource security pressure will gradually diminish. In contrast, the medical resource security pressure will increase in the surrounding cities that absorb population. Therefore, it is necessary to pay more attention to the population growth in the region and promote the implementation of policies and the layout of the medical industry in the corresponding areas.

6. Conclusion

In this paper, we constructed a set of regional resilience assessment framework for medical resources security under the operation of medical resource synergy, which can achieve the results of the resilience impacts of each city nodes, the resilience measurement of network structural and the resilience ability of recovery under attacks. By collecting data on the supply and demand of medical resources in YRD, PRD and BTH, we calculated the synergistic intensity of medical resource among the cities in the three city clusters and performed network modelling. The results show that there are leading cities for medical resource synergy within each city cluster. In general, the regional medical resource synergy strength between in-cities in PRD is the best, while in YRD is the most balanced. Beijing shows the greatest medical impact in BTH. By comparing the development of regional medical resource security resilience under the time periods of 2010, 2015 and 2020, PRD and BTH show the trend of a high medical concentration in the core cities, while YRD region exhibits a more balanced pattern of development in multiple sub-regions.

However, based on the existing research foundation, this study has some limitations. Firstly, due to the limited nature of the data, this study only considers the synergistic relationship of medical resources at the prefecture-level city scale. With the development of integration within the city cluster, healthcare interactions within smaller regional scales are more frequent, and people's demand for healthcare coverage in close proximity is more urgent. Secondly, In this study, only two

indicators—Number of Beds of Hospitals and Number of Licensed (Assistant) Doctors, were selected for the broad concept of regional medical resources. In China, hospitals of different types and grades exist within cities. Among them, Grade-A tertiary hospitals assume the main body of cross-city resource synergy activities within the region, which is also an important influencing factor. Further consideration of the impact of these indicators on the medical resource synergy security resilience will be an important direction for the study to continue to deepen. Last but not least, regional policies and management will also have a significant impact on regional medical resource security, with policy directions affecting the distribution of healthcare resources and the synergistic development between cities. Furthermore, for the quantitative analysis of the degree of security under the assessment of regional medical resources supply and demand, there are significant differences in the degree of demand among different population groups within a city. Therefore, our future research work will further integrate the regional medical policies of city cluster and the medical resources needs of different populations to study this issue in depth.

CRedit authorship contribution statement

Longbin Kou: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hanping Zhao:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Zhixiang Yang:** Resources, Methodology. **Xianghui Li:** Methodology, Data curation. **Yichi Zhang:** Data curation. **Jinfan Liang:** Software. **Haoyue Qiu:** Data curation. **Yumian Zhang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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